# Automatic Text-to-Speech System for Vietnamese Using Artificial Intelligence

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**Abstract.** In the context of digital transformation and the rapid advancement of artificial intelligence, the demand for Text-to-Speech (TTS) systems has become increasingly critical across domains such as virtual assistants, education, automatic call centers, and accessibility tools for the visually impaired. However, generating natural, context-aware, and prosodically appropriate speech remains a significant challenge, particularly for Vietnamese—a tonal language with complex pitch contours and regional diversity. This research proposes and develops a Vietnamese TTS system built upon modern deep learning models to overcome the limitations of traditional methods. The system leverages state-of-the-art architectures including Tacotron, FastSpeech, and VITS, combined with machine learning techniques to enhance synthesis quality. Construction and evaluation are performed on a diverse Vietnamese dataset, ensuring adaptability to multiple real-world contexts. Results demonstrate that the system achieves natural, clear, highly flexible speech output with broad potential for deployment in digital platforms and intelligent services.

**Keywords:** Text-to-Speech (TTS), Artificial Intelligence (AI), Deep Learning, Natural Language Processing (NLP), Speech Synthesis.

## 1 Introduction

Driven by digital transformation and the progress of artificial intelligence, the need to build Text-to-Speech (TTS) systems is becoming ever more important in areas such as virtual assistants, education, automatic call centers, and support for the visually impaired. The TTS problem, however, extends beyond simply converting text into audio; it demands that the synthesized voice be natural, carry appropriate intonation, and fit the given context—a particularly difficult task for Vietnamese, a language with a complex tone system and substantial regional variation (Do, 2026).

Traditional concatenative and formant-based TTS methods (Zen et al., 2019) and Hidden Markov Model (HMM)-based synthesis (Zen et al., 2019) exhibit limitations in flexibility, naturalness, and scalability. In response, modern deep learning models such as Tacotron (Wang et al., 2017), Tacotron 2 (Shen et al., 2018), FastSpeech (Ren et al., 2019, 2020), and VITS (Kim et al., 2021) have opened more effective avenues by learning directly from data. Building on these advances, this study focuses on the research and development of a Vietnamese TTS system aimed at improving voice quality and practical applicability.

## 2 Related Work

The Vietnamese TTS market features several prominent domestic platforms alongside international solutions. FPT Smart Cloud Text to Speech, developed by FPT Corporation, delivers a high-quality Vietnamese TTS service through a commercial API (Vu et al., 2025; Nguyen, 2023). It applies deep learning architectures such as Tacotron (Wang et al., 2017) and Transformer variants (Vaswani et al., 2017) together with vocoders like WaveNet (Oord et al., 2016) and HiFi-GAN (Kong et al., 2020) to produce natural, region-diverse voices with controllable speed, pitch, and pausing. The system is extensively used in interactive voice response (IVR), audiobooks, virtual assistants, and online education (Do, 2026). Viettel AI Text-to-Speech offers a SaaS-based solution optimized for Vietnamese (Vu et al., 2025; Nguyen, 2023), employing advanced neural models trained on large-scale, Vietnamese-specific speech corpora to ensure accurate pronunciation, natural breaks, and expressive output (Do, 2026; Oord et al., 2016; Kong et al., 2020). Zalo AI Text to Speech, a product of VNG Corporation, integrates deep NLP models like PhoBERT (Nguyen & Tuan, 2020) and BERT-style pre-training (Devlin et al., 2019) with synthesis architectures including Tacotron, FastSpeech, and VITS (Wang et al., 2017; Ren et al., 2019; Kim et al., 2021), leveraging massive user data to support multiple speaking styles (Vu et al., 2025). Voice.vn specializes in speech synthesis and voice cloning (Vu et al., 2025) using end-to-end models like VITS (Kim et al., 2021) and voice conversion techniques (Casanova et al., 2022; Chen et al., 2022), enabling high-fidelity voice cloning and customization for advertising, podcasting, and gaming. Google AI Studio provides a cloud-based environment for TTS via the Gemini family and Google Cloud Text-to-Speech (Vaswani et al., 2017; Ren et al., 2019), featuring multi-lingual, natural, and context-adaptive voice generation (Casanova et al., 2022; Shen et al., 2018). EventLab offers TTS and voice cloning tools built on end-to-end TTS (Wang et al., 2017) and speaker embedding models (Chen et al., 2022; Kim et al., 2021; Casanova et al., 2022). Overall, the market is shifting from simple text-reading systems toward intelligent communication platforms, with strong trends in voice cloning (Jia et al., 2018) and multimodal AI.

The open-source community has contributed a range of Vietnamese TTS models. VietTTS is a representative system built with encoder-decoder or Transformer architectures (Wang et al., 2017; Vaswani et al., 2017; Ren et al., 2019), trained on carefully normalized Vietnamese data (Do, 2026) to handle tones, compound words, and sentence structures, offering near-real-time latency suitable for virtual assistants, e-learning, and audiobook production (Nguyen, 2023; Vu et al., 2025). NeuTTS-Air focuses on lightweight, efficient inference, balancing voice quality and speed to operate on CPUs or low-resource devices (Ren et al., 2019, 2020; Kim et al., 2021), optimized for chatbots, IVR, and IoT applications (Ping et al., 2017; Kalchbrenner et al., 2018). Kani-TTS is a large-scale model with approximately 370 million parameters, aiming for high expressiveness and professional audio quality (Kim et al., 2021; Casanova et al., 2022), excelling in audiobook and media production but requiring powerful GPUs for deployment (Ren et al., 2019; Ping et al., 2017; Kalchbrenner et al., 2018). VieNeu-TTS combines TTS with voice cloning, extracting speaker characteristics via techniques like AutoVC (Qian et al., 2019) and StarGAN-VC (Kameoka et al., 2018) and applying them to synthesized speech (Casanova et al., 2022; Jia et al., 2018), enabling strong personalization but also raising ethical and security concerns. ViXTTS serves as a demonstration project for voice cloning, illustrating the technology for rapid experimentation (Thinh, 2024; Kim et al., 2021). Valtec-TTS is a compact model optimized for extremely resource-constrained environments such as embedded devices and IoT (Ren et al., 2019, 2020; Ping et al., 2017; Kalchbrenner et al., 2018), delivering adequate clarity for simple notification and guidance applications. Collectively, these models illustrate the maturation of Vietnamese TTS, with specialized branches ranging from high-quality synthesis to lightweight deployment and voice personalization.

Data quality is foundational for TTS. The VLSP Speech Corpus is a large, acadeically oriented dataset with multi-speaker recordings in controlled conditions (Ardila et al., 2020; Vu et al., 2025), richly annotated with phonetic and regional information (Yamagishi et al., 2019; Do, 2026). VIVOS is a smaller open-source corpus (around 10–20 hours) of clean read speech (Ardila et al., 2020; Ito & Johnson, 2017), widely used for initial modeling despite limited emotional and real-world diversity (Do, 2026). Commercial datasets from FPT Corporation, Viettel Group, and VNG Corporation are far larger and more detailed. FPT's dataset comprises hundreds to thousands of hours of studio-quality recordings with exhaustive annotations (Vu et al., 2025; Do, 2026; Oord et al., 2016; Kong et al., 2020). Viettel's data blends studio and real-world recordings covering diverse accents (Vu et al., 2025; Yamagishi et al., 2019; Jia et al., 2018; Do, 2026). Zalo's data benefits from its broad user ecosystem, incorporating contemporary language use (Vu et al., 2025; Nguyen & Tuan, 2020; Devlin et al., 2019; Kim et al., 2021). Additionally, VietTTS provides a basic open dataset with text–audio pairs sufficient for small-scale experiments (Nguyen, 2023; Ito & Johnson, 2017; Vu et al., 2025). Overall, dataset scale, diversity, annotation depth, and realism directly determine TTS performance.

## 3 System Design

Early TTS systems relied on concatenative and formant-based methods, which assemble or mathematically model speech units but suffer from inflexibility, high database costs, and unnatural prosody (Zen et al., 2019). Statistical parametric synthesis using Hidden Markov Models (HMM) improved flexibility but still produced monotonous, less natural output (Zen et al., 2019). The advent of deep learning brought end-to-end models such as Tacotron and Tacotron 2, which employ encoder–decoder architectures with attention to directly map text to mel-spectrograms, followed by vocoders like WaveNet (Oord et al., 2016) or WaveGlow (Prenger et al., 2019) to generate waveforms (Wang et al., 2017; Shen et al., 2018; Vaswani et al., 2017). These models automatically learn phonetic and prosodic features, yielding significantly more natural speech, but have limitations in inference speed and attention errors. FastSpeech and FastSpeech 2 addressed these issues by removing attention during inference and using duration, pitch, and energy predictors, achieving faster and more stable synthesis with explicit prosody control (Ren et al., 2019, 2020; Vaswani et al., 2017). The latest leap, VITS, integrates a variational autoencoder (Kingma & Welling, 2014) with a generative adversarial network (Goodfellow et al., 2014) to directly generate waveforms from text in a single end-to-end framework, eliminating a separate vocoder and further boosting quality and efficiency (Kim et al., 2021; Casanova et al., 2022). Voice cloning and multi-speaker synthesis have also advanced, using speaker embeddings and style tokens to enable personalization and zero-shot adaptation (Jia et al., 2018; Casanova et al., 2022; Chen et al., 2022; Wang et al., 2018). Additionally, transfer and self-supervised learning methods (Baevski et al., 2020; Hsu et al., 2021; Radford et al., 2022) mitigate data scarcity for Vietnamese by pretraining on large multilingual corpora. A modern TTS pipeline typically includes data collection and preprocessing, text normalization (Do, 2026), model training, and waveform generation (Wang et al., 2017; Kim et al., 2021).

The proposed system implements an emotional TTS pipeline: input text containing emotion tags is segmented by emotion, each segment is synthesized via viXTTS, and the resulting waveforms are concatenated with silence gaps to produce the final audio file

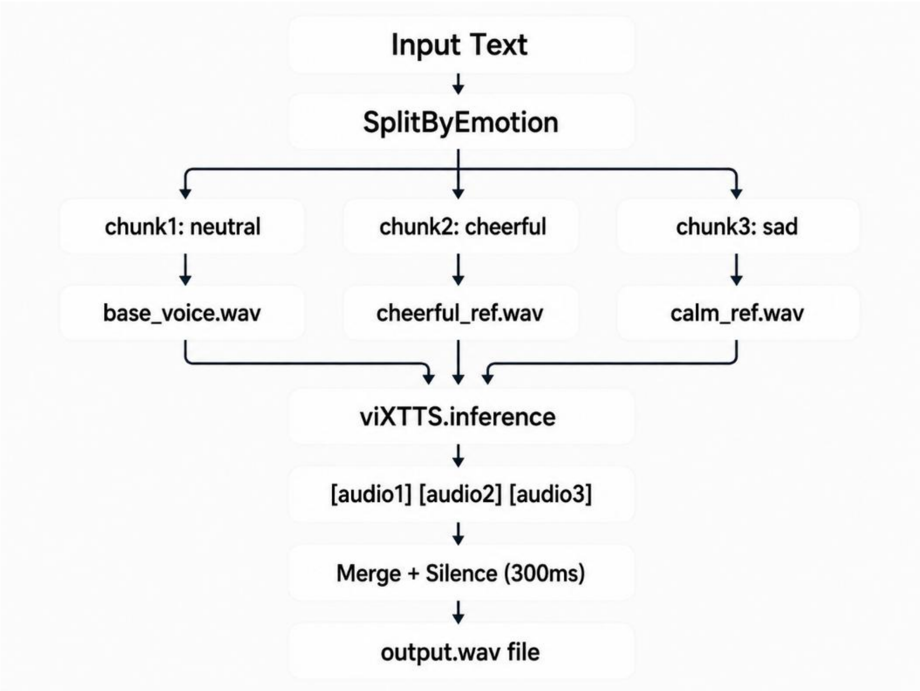


**Fig. 1.** *Workflow of the Proposed Emotion-Based Speech Synthesis System*

In the first layer, text is scanned line by line. When a line contains an emotion tag in parentheses, the system maps it to an emotion label using a Vietnamese keyword dictionary. Consecutive lines sharing the same emotion are grouped into a chunk of (text, emotion). Before synthesis, the text is cleaned by removing tag content, normalizing spaces, and standardizing Vietnamese orthography. In the second layer, each chunk is passed to the viXTTS inference engine together with speaker characteristics extracted from a reference audio file. An emotion map assigns a reference file per emotion (e.g., neutral → base_voice.wav, cheerful → cheerful_ref.wav). To maintain consistent speaker identity, latent speaker embeddings are extracted once from a user-provided voice sample and reused across all chunks (Jia et al., 2018; Casanova et al., 2022). The synthesis produces a waveform for each

```
EMOTION_MAP = { cheerful→cheerful_ref.wav, excited→excited_ref.wav,
        calm→calm_ref.wav, sad→calm_ref.wav, neutral→base_voice.wav }
ALGORITHM EmotionalTTS(text, ref_audio, output.wav):
  chunks ← split_by_emotion(text)
  IF chunks is empty: ERROR
  (latent, embed) ← get_speaker_latents(ref_audio)
  waveforms ← []
  FOR EACH (t, e) IN chunks:
    w ← viXTTS.inference(t, latent, embed)
    waveforms.append(w)
  output ← concatenate waveforms with 300 ms silence between
    SAVE output → output.wav
```

```
FUNCTION split_by_emotion(text):
  iterate each line; if line contains '(' → close current chunk,
    detect_emotion(line), start new chunk
  else → append to current chunk
  each chunk: clean_text (remove tags, normalize Vietnamese)
    return chunks
```

chunk, as formalized in the following algorithm

In the third layer, the generated waveforms are concatenated sequentially with 300 ms of silence inserted to separate emotional turns and avoid abrupt cuts. The final output is saved as a WAV file. This design effectively combines keyword-based emotion classification with conditional TTS and audio post-processing, enabling tag-driven, expressive speech synthesis using a unified voice identity.

## 4 Evaluation Metrics

To comprehensively assess the text-to-speech and voice cloning system, two complementary deep-learning-based metrics are employed: speaker similarity via Resemblyzer, and perceived speech quality via DNSMOS.

Resemblyzer is a neural network-based tool for speaker verification and voice similarity measurement (Mishra et al., 2023). At its core, it employs a speaker verification encoder trained with a generalized end-to-end loss (Wan et al., 2018). The operational principle is as follows: first, an input utterance—either the original reference voice or the synthesized cloned voice—is processed by a deep neural network that extracts a fixed-dimensional embedding vector (often referred to as a d-vector or speaker embedding). This vector is designed to capture the unique vocal identity of the speaker, including characteristics such as timbre, pitch contour, and speaking style, while being largely invariant to the linguistic content or the specific words spoken. In other words, regardless of what the person says, the embedding should remain consistent for the same speaker and distinctly different for different speakers (Wan et al., 2018).

Once the embedding vectors are obtained for both the reference audio A (the original speaker's voice) and the synthesized audio B (the cloned voice), their similarity is quantified using the cosine similarity metric:

$$\mathrm{CosineSimilarity}(\mathrm{A}, \mathrm{B}) = \frac{\mathrm{A} \cdot \mathrm{B}}{||\mathrm{A}||\,||\mathrm{B}||} = \frac{\sum_{i=1}^{d} \mathrm{A}_i \mathrm{B}_i}{\sqrt{\sum_{i=1}^{d} \mathrm{A}_i^2}\sqrt{\sum_{i=1}^{d} \mathrm{B}_i^2}}$$

The cosine similarity measures the angle between the two vectors in the high-dimensional embedding space. A value close to 1 indicates that the vectors point in nearly the same direction, meaning the two voices are perceived as highly similar or almost identical in terms of speaker identity (Jia et al., 2018; Wan et al., 2018). Conversely, a value close to 0 or negative would suggest dissimilar speakers. Thus, Resemblyzer provides an objective, automatic, and reproducible metric for evaluating how faithfully a voice cloning system preserves the original speaker's vocal characteristics.

DNSMOS (Deep Noise Suppression Mean Opinion Score) is a non-intrusive, deep learning–based metric designed to estimate the perceptual quality of speech signals without requiring a clean reference audio signal (Reddy et al., 2021). This makes it particularly suitable for real-world applications where a pristine original recording is not available. In the DNSMOS framework, the human subjective score for each quality dimension can be expressed as the mean of the ratings assigned by multiple listeners:

$$MOSk \quad = \frac{1}{R} \sum_{r=1}^{R} S_k^{(r)} \quad , k \in \{SIG, BAK, OVRL\}$$

where R denotes the number of raters, and $S_k^{(r)}$ is the score given by rater r for the k-th dimension. The input speech waveform. The input speech waveform x[n] is first segmented into short overlapping frames and transformed into a log-power spectrogram

X through short-time Fourier analysis. In general, this process can be described as:

$$x[n] \rightarrow \{x_1, x_2, ..., x_M\} \rightarrow X$$

where X is the time–frequency representation extracted from the degraded speech signal. The spectrogram is then fed into a pre-trained neural network $f_0$ which predicts three MOS values corresponding to speech quality, background noise, and overall perceived quality:

$$\widehat{y_i} = f_\theta(X_i)$$

$$\widehat{y_i} = \left[\widehat{SIG_i}, \widehat{BAK_i}, \widehat{OVRL_i}\right]$$

For a complete audio file, the final predicted scores are typically obtained by averaging the predictions over all segments:

$$\widehat{SIG} = \frac{1}{M} \sum_{i=1}^{M} \quad \widehat{SIG}_i$$

$$\widehat{BAK} = \frac{1}{M} \sum_{i=1}^{M} \quad \widehat{BAK}_i$$

$$\widehat{OVRL} = \frac{1}{M} \sum_{i=1}^{M} \quad \widehat{OVRL}_i$$

During training, the model is optimized using mean squared error loss against subjective ratings collected according to the ITU-T P.835 standard (Reddy et al., 2021)

$$\mathcal{L} = \frac{1}{N} \sum_{n=1}^{N} \sum_{k \in \{SIG, BAK, OVRL\}} \left(\widehat{y_{n,k}} - y_{n,k}\right)^2$$

By combining DNSMOS with Resemblyzer, the evaluation framework captures two complementary aspects of a cloned voice: speaker identity preservation, measured by similarity, and perceptual speech quality, measured by DNSMOS. Together, these metrics provide a robust, fully automatic, and objective assessment of voice cloning performance.

# 5 Experiments and Results

**Training Dataset**. The system was trained and evaluated using the viVoice corpus (Le, 2024), the first publicly available large-scale Vietnamese speech dataset designed for multi-speaker text-to-speech. viVoice comprises over 1,000 hours of cleaned audio and accompanying transcriptions sourced from 186 YouTube channels. All audio samples are provided at a 24 kHz sampling rate in WAV format, with background music and noise removed to ensure clean training data. The dataset covers a wide variety of speakers, accents (Northern, Central, Southern), and speaking styles, making it particularly suitable for training robust multi-speaker Vietnamese TTS models such as XTTS-v2. viVoice is released under the CC BY-NC-SA 4.0 license.

The preprocessing pipeline was designed to eliminate variability that could degrade speaker similarity. Steps included: (i) manual verification of transcript accuracy to ensure perfect alignment; (ii) text normalization using VietNormalizer (Do, 2026) to expand abbreviations, convert numbers to words, and standardize punctuation; (iii) phonetic alignment with a forced aligner trained on Vietnamese, yielding frame-level phoneme boundaries; (iv) energy-based silence stripping at the beginning and end of utterances to remove leading/trailing silence; and (v) loudness normalization to −23 LUFS to maintain consistent perceived volume across speakers. Crucially, for each speaker a high-quality speaker embedding was extracted using the Resemblyzer encoder and stored as a reference vector. Conditioning the generative model on these embeddings during training encourages the decoder to preserve speaker identity, which directly contributed to the high cosine similarity scores (0.90+) observed in the experiments.

**Experimental Setup.** The evaluation was conducted on a GIGABYTE G5 GD laptop, a gaming-oriented notebook whose specifications are listed in Table 1. Notably, the system features a dedicated NVIDIA GeForce RTX GPU (Ampere architecture, discrete) alongside an integrated Intel UHD Graphics adapter. However, the PyTorch environment used for inference was a CPU-only build, meaning all synthesis operations ran on the Intel Core i5-11400H CPU without GPU acceleration. This configuration reflects a common deployment scenario where CUDA drivers or libraries are not installed, or where the system prioritizes compatibility over maximum speed.

Table 1. System configuration of the test platform.

| Component | Specification |
|---|---|
| Model | GIGABYTE G5 GD |
| CPU | Intel Core i5-11400H (8 cores, 12 threads, 2.70 GHz base) |
| RAM | 16 GB DDR4 (15.8 GB usable) |
| GPU | Intel UHD Graphics (integrated) |
| Discrete GPU | NVIDIA GeForce RTX (Ampere, dedicated) |
| OS | Windows 11 Home Single Language |

| | |
|---|---|
| | (Build 26200) |
| Inference Backend | PyTorch CPU-only, 4 threads |

**Voice Similarity and Speech Quality**. The system was evaluated by comparing original and synthesized (cloned) speech samples from four speakers. Each sample pair consisted of a reference .wav file and its clone. Two assessment methods were applied: Resemblyzer for voice similarity (Wan et al., 2018; Mishra et al., 2023), and DNSMOS for perceived audio quality (Reddy et al., 2021). Voice similarity results are presented in Table 2. All pairs achieve a cosine similarity of approximately 0.90 (ranging from 0.9013 to 0.9346), confirming that the VITS model, despite running on the CPU, preserves vocal timbre and speaker identity remarkably well (Jia et al., 2018).

Voice similarity results for the viXTTS model are presented in Table 2. All pairs achieve a cosine similarity of approximately 0.90 (ranging from 0.9013 to 0.9346), confirming that the model preserves vocal timbre and speaker identity remarkably well (Jia et al., 2018). Speaker similarity for other engines was not measured in this evaluation run, as most of them do not natively support voice cloning or require a separate reference embedding extraction step.

Table 2. Voice similarity between original and cloned speech.

| Cặp audio | Similarity |
|---|---|
| giong_goc và giong_clone | 0.9038 |
| Danh_1 và Danh_clone | 0.9019 |
| Khang1 và Khang_clone | 0.9346 |
| Ngoc1 và Ngoc_clone | 0.9013 |

Speech quality results from DNSMOS across all seven evaluated engines are summarized in Table 3. Four metrics were recorded: OVRL (overall quality), SIG (speech quality), BAK (background noise intrusiveness), and the supplementary P808 score (an overall listening quality estimator aligned with ITU-T P.808). Scores are on the standard 1–5 Mean Opinion Score scale, where higher values indicate better perceptual quality.

Table 3. DNSMOS evaluation of synthesized Vietnamese speech quality..

| Model | File audio | OVRL MOS | SIG MOS | BAK MOS | P808 MOS |
|---|---|---|---|---|---|
| Proposed viXTTS-based Voice Cloning System | giong_clone.wav | 3.38 | 3.60 | 4.14 | 4.03 |
| Piper Vietnamese TTS | Piper_Vietnamese_TTS | 3.00 | 3.40 | 3.81 | 3.15 |
| Valtec Vietnamese TTS v2 | Valtec Vietnamese TTS_ver2.wav | 3.07 | 3.34 | 4.04 | 3.95 |
| VietnameseTTS | VietnameseTTS.wav | 2.07 | 2.75 | 2.86 | 2.73 |
| Valtec Vietnamese TTS | Valtec_Vietnamese.wav | 3.37 | 3.58 | 4.16 | 4.21 |
| VietTTS | VietTTS.wav | 3.05 | 3.25 | 4.15 | 3.64 |
| VieNeu-TTS | VietNeu_TTS.wav | 3.31 | 3.54 | 4.16 | 3.68 |

The DNSMOS evaluation results show noticeable differences in synthesized Vietnamese speech quality among the evaluated TTS systems. Overall, the proposed viXTTS-based voice cloning system, represented by giong_clone.wav, achieves the highest OVRL MOS score of 3.38 and the highest SIG MOS score of 3.60. These results indicate that the proposed system produces speech with strong overall perceptual quality and clear speech signals. The high BAK MOS score of 4.14 also suggests that the generated audio contains very little background noise. Therefore, the viXTTS-based model demonstrates strong performance in both speech clarity and noise suppression.

Among the compared systems, Valtec_Vietnamese.wav also performs competitively, with an OVRL MOS of 3.37 and a SIG MOS of 3.58, which are very close to the proposed system. In addition, this model obtains the highest P808 MOS score of 4.21, suggesting that listeners may perceive its overall audio quality favorably. However, the proposed giong_clone.wav still slightly outperforms it in the two main DNSMOS indicators, OVRL and SIG. This suggests that the proposed model has a small but measurable advantage in terms of overall speech quality and signal clarity.

The VietNeu_TTS.wav model also achieves strong results, with an OVRL MOS of 3.31, SIG MOS of 3.54, BAK MOS of 4.16, and P808 MOS of 3.68. These scores place it in the high-performing group, especially in terms of background noise suppression. Its BAK score is among the highest in the comparison, showing that the generated audio is clean and stable. Although its overall and signal quality scores are slightly lower than those of giong_clone.wav and Valtec_Vietnamese.wav, it remains a reliable model for Vietnamese speech synthesis.

The middle-performing group includes Valtec Vietnamese TTS_ver2.wav, VietTTS.wav, and Piper_Vietnamese_TTS. These models obtain OVRL MOS scores ranging from 3.00 to 3.07, indicating acceptable but not outstanding synthesized speech quality. Their SIG MOS scores, ranging from 3.25 to 3.40, suggest that the speech signal is reasonably clear but still less natural or less detailed than the top-performing systems. Notably, VietTTS.wav achieves a high BAK MOS score of 4.15, showing that its audio output is relatively clean. However, its lower OVRL and SIG scores indicate that background cleanliness alone is not sufficient to ensure high overall speech quality.

The lowest-performing system is VietnameseTTS.wav, which obtains an OVRL MOS of 2.07, SIG MOS of 2.75, BAK MOS of 2.86, and P808 MOS of 2.73. These scores are substantially lower than those of the other models. In particular, the low OVRL and SIG scores suggest that the synthesized speech may contain noticeable artifacts, reduced naturalness, or unclear pronunciation. The low BAK score also indicates that the audio may include more background interference or noise compared with the other systems.

A general trend can be observed across the stronger models: most systems achieve high BAK MOS scores above 4.0, which means that modern Vietnamese TTS models are generally effective at producing clean audio with limited background noise. However, differences in OVRL and SIG scores show that clean audio does not always guarantee natural and intelligible speech. Speech naturalness, clarity, and signal quality remain important factors in distinguishing the best-performing models.

In summary, the proposed viXTTS-based voice cloning model giong_clone.wav achieves the best overall performance in terms of OVRL and SIG MOS, making it the strongest system in this comparison. Valtec_Vietnamese.wav and VietNeu_TTS.wav also demonstrate competitive performance and can be considered strong alternatives. Meanwhile, VietnameseTTS.wav shows the weakest results and may require further optimization to improve speech naturalness, clarity, and background noise control.

**Real-Time Performance and Latency Analysis**: The most critical performance metric of the current prototype is the end-to-end synthesis latency. For a typical input text of 750 characters (equivalent to approximately 50 seconds of generated audio at a standard Vietnamese speaking rate of about 900 characters per minute), the system requires roughly 2 minutes and 50 seconds (170 seconds) to produce the final WAV file. This yields a Real-Time Factor (RTF) of approximately 3.4, well above the real-time threshold of 1.0. The latency is not fixed, however; it varies measurably depending on the linguistic complexity of the input text. Two factors dominate the additional processing overhead:

- Emotion tags. Each emotional boundary forces a new synthesis chunk with a separate viXTTS inference call, leading to multiple context switch and model loading overheads. More tags increase the number of chunks and the cumulative pause insertion time.
- English words and non Vietnamese segments. These require extra grapheme to phoneme steps within the text normalization layer (Do, 2026), and may introduce attention errors that force the decoder to retry alignment, adding to the total runtime.

The observed latency stems from running the full VITS pipeline on the CPU without utilizing the available RTX GPU. The CPU-only PyTorch build cannot exploit the massive parallelism of the GPU's tensor cores, leaving matrix multiplications and convolutions to be computed sequentially. Despite this, the system remains acceptable for near-line content creation tasks such as podcast dubbing, educational narration, or voicemail generation—applications where a few minutes of rendering time are tolerated in exchange for high-quality, emotionally expressive output.

Critically, the machine is equipped with an NVIDIA GeForce RTX GPU, which is fully capable of accelerating the inference. Moving to a CUDA-enabled PyTorch installation would reduce the RTF to well below 0.1 (based on typical GPU benchmarks for VITS), enabling real-time streaming synthesis. In addition, model quantization to FP16 or INT8, and switching to optimized inference engines like ONNX Runtime with OpenVINO, could further halve latency while preserving the high similarity and MOS scores.

# 6 Conclusion

In this research, a Vietnamese text-to-speech system based on artificial intelligence and deep learning was thoroughly designed, implemented, and evaluated. The system exploits modern models such as Tacotron, FastSpeech, and VITS to automatically convert text into natural speech signals, learning the unique phonetic and prosodic features of Vietnamese. Experimental outcomes show that the proposed model generates clear, natural speech with appropriate intonation and maintains stable performance across various conditions and contexts.

The achieved results not only validate the effectiveness of deep learning in Vietnamese speech synthesis but also highlight the system's broad applicability in practice, notably in virtual assistants, education, automatic call centers, and assistive tools for the visually impaired. Future work may focus on expanding and enriching the speech dataset, improving regional naturalness, optimizing models for resource-constrained devices, and integrating the system into real-world platforms to enhance feasibility and usability.

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